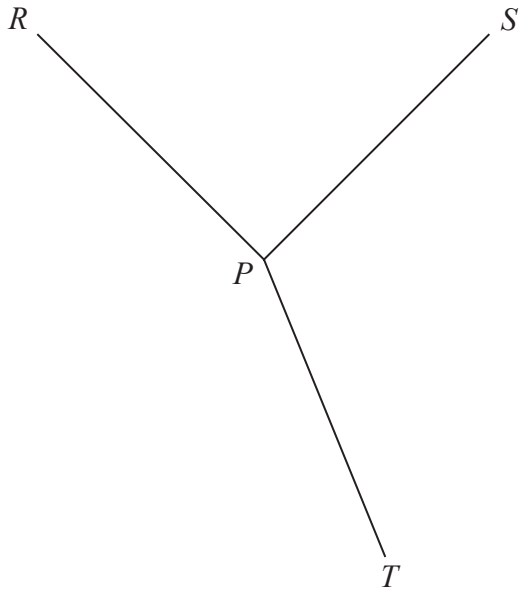
9



PR and *PS* are two sides of a square.
PS and *PT* are two sides of a regular pentagon.

PR and *PT* are two sides of a regular polygon with *n* sides.

Work out the value of *n*.
You must show your working.

Pentagon: (5 – 2) × 180° ÷ 5 = 108°

1 mark
Pentagon interior angle.

At P: 360° – 90° – 108° = 162°

1 mark
Angles at P total 360°.

Exterior angle = 180° – 162° = 18°

1 mark
Use the exterior angle for n.

Number of sides = 360° ÷ 18° = 20

1 mark
20 sides.

n = ...

(Total for Question 9 is 4 marks)